

Emacs Configuration Notes & Refactoring Plan

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1 DistroTube's Emacs Configuration Analysis

1.1 Overview

Derek Taylor's (DistroTube) Emacs configuration is a literate programming setup using org-babel. The entire configuration lives in a single `config.org` file (~1,161 lines) that tangles to `config.el`.

Repository: <https://gitlab.com/dwt1/dotfiles> (`.config/emacs/`)

1.2 File Structure

```
.config/emacs/
├── init.el                # 4 lines - just loads config.org
├── early-init.el         # Disables package.el
├── config.org             # Main literate config (1,161 lines)
├── config.el             # Auto-generated from config.org
├── start.org             # Help/reference document
├── scripts/              # Modular elisp helpers
│   ├── elpaca-setup.el   # Package manager bootstrap
│   ├── app-launchers.el  # dmenu-like launcher
│   ├── buffer-move.el    # Window movement utilities
│   └── eshell-prompt.el  # Custom eshell prompt with git
├── themes/
│   └── dtmacs-theme.el   # Custom theme
├── eshell/
│   ├── aliases           # Shell aliases
│   └── profile           # Startup commands
└── images/
    └── dtmacs-logo.png   # Dashboard logo
```

1.3 Org-Babel Tangling Pattern

1.3.1 The init.el Loader

The entire `init.el` is just:

```
(org-babel-load-file
  (expand-file-name "config.org" user-emacs-directory))
```

1.3.2 config.org Header

```
#+TITLE: DT's GNU Emacs Config
```

```
#+AUTHOR: Derek Taylor (DT)
#+STARTUP: showeverything
#+OPTIONS: toc:2
```

1.3.3 Code Block Pattern

Each section contains prose documentation followed by code:

```
* SECTION NAME
Description of what this section does and why.
```

```
#+begin_src emacs-lisp
;; Actual elisp code here
#+end_src
```

1.3.4 Scripts Directory Pattern

Large standalone utilities go in `scripts/` and are loaded via:

```
(add-to-list 'load-path "~/.config/emacs/scripts/")
(require 'elpaca-setup)
(require 'buffer-move)
(require 'app-launchers)
```

1.4 Key Packages

Category	Packages	Notes
Package Manager	Elpaca	Async, modern (you use straight.el)
Keybindings	Evil, general	Vim-style with SPC leader
Completion	Ivy, Counsel	(you use Helm)
UI	dashboard, doom-modeline, doom-themes	Rich startup
Git	Magit, Forge, git-timemachine	Full git workflow
Org	org-bullets, toc-org, ox-hugo	Similar to yours
File Tree	Neotree	Sidebar browser
Workspaces	Perspective	Named buffer groups
AI	Ellama	Local LLM via Ollama
Media	Elfeed (RSS), Eradio (radio)	Entertainment

1.5 Config.org Section Breakdown

Section	Lines	Purpose
Important Programs	14	Load scripts
All The Icons	10	Icon setup
Backup	6	Trash-based backups
Company	19	Completion
Dashboard	32	Startup screen
Evil	31	Vim keybindings
Fonts	52	JetBrains Mono + Ubuntu
General Keybindings	228	SPC leader map
Git Programs	41	Magit/Forge
Ivy	39	Completion framework
Org Mode	88	Core org settings
Shells/Terminals	85	Eshell, Vterm
Theme	14	doom-one
Which-key	19	Keybinding hints

2 Features of Interest

2.1 Dashboard (Startup Screen)

The `dashboard` package provides a rich startup buffer with quick access to:

- Recent files
- Agenda items
- Bookmarks
- Projects

2.1.1 Installation

```
(use-package dashboard
  :straight t
  :init
  (setq initial-buffer-choice 'dashboard-open
        dashboard-set-heading-icons t
        dashboard-set-file-icons t
        dashboard-banner-logo-title "Welcome, Aayush!"
        dashboard-startup-banner 'logo ; or path to image
```

```

dashboard-center-content nil
dashboard-items '((recents . 5)
                  (agenda . 5)
                  (bookmarks . 3)
                  (projects . 3)))

:config
(dashboard-setup-startup-hook))

```

2.1.2 Comparison with my-home.el

Feature	Dashboard	my-home.el
Recent files	Built-in	Built-in
Agenda	Built-in	Not present
Bookmarks	Built-in	Not present
Projects	Built-in (projectile)	Not present
Custom logo	Supported	Not present
Customization	Extensive options	Full control

Decision: You already have `my-home.el` with custom navigation. Could enhance it with agenda/bookmarks, or switch to dashboard for consistency with distrotube pattern.

2.2 Perspective (Workspaces)

The `perspective` package provides named workspaces with isolated buffer lists.

2.2.1 Key Features

- Each perspective has its own buffer list
- Switch between perspectives by name
- Sessions persist across restarts
- Integrates with `ibuffer` for filtering

2.2.2 Installation

```

(use-package perspective
  :straight t
  :custom

```

```
(persp-mode-prefix-key (kbd "C-c M-p"))
:init
(persp-mode)
:config
(setq persp-state-default-file
  (expand-file-name "sessions" user-emacs-directory)))
```

2.2.3 Distrotube's Keybindings

Key	Command	Description
C-c M-p s	persp-switch	Switch/create perspective
C-c M-p k	persp-remove-buffer	Remove buffer from perspective
C-c M-p c	persp-kill	Kill a perspective
C-c M-p r	persp-rename	Rename perspective
C-c M-p a	persp-add-buffer	Add buffer to perspective
C-c M-p A	persp-set-buffer	Set buffer to only this perspective
C-c M-p b	persp-switch-to-buffer	Switch buffer within perspective
C-c M-p i	persp-import	Import buffers from another perspective
C-c M-p n/p	persp-next/prev	Cycle perspectives

2.2.4 Use Cases

- Separate workspaces for: email, coding, writing, org-roam
- Quick context switching without buffer clutter
- Save/restore workspace layouts

3 Your Current Configuration Summary

3.1 File Structure (15 modules, ~5,648 lines)

```
~/.emacs.d/
├── init.el                # Orchestrator
├── early-init.el          # GC + native-comp (macOS ARM)
├── elisp/
│   ├── bootstrap.el      # straight.el setup
│   ├── package-config.el # 2,191 lines - Core packages
│   ├── org-config.el     # 1,338 lines - Org + LaTeX
│   └── email-config.el   # 748 lines - notmuch
```

		anki-config.el	# 236 lines - Anki-editor
		my-home.el	# 234 lines - Custom startup
		gruber-themes.el	# 168 lines - Theme toggle
		java-lsp.el	# 131 lines - Java LSP
		auto-save-config.el	# 80 lines - Auto-save
		ui-config.el	# 79 lines - Theme/fonts
		custom-vars.el	# 59 lines - Customize
		ox-hugo-bindings.el	# 49 lines - Hugo export
		magit-bindings.el	# 27 lines - Git keys
		ink.el	# 236 lines - Inkscape
		ob-markdown.el	# 32 lines - Pandoc babel
		custom-themes/	# Gruber theme variants
		straight/	# Package manager data
		backups/, auto-save/	# Runtime directories

3.2 Key Differences from Distrotube

Aspect	Your Config	Distrotube
Structure	15 modular .el files	1 config.org file
Lines	~5,648	~1,161
Package Manager	straight.el	Elpaca
Completion	Helm	Ivy/Counsel
Keybindings	Emacs-style + custom	Evil (Vim) + general.el
Email	notmuch (3 accounts)	Not configured
L ^A T _E X	Advanced parallel preview	Basic
Org-roam	Extensive dailies system	Not configured
Anki	Full integration	Not present
Hugo	Bidirectional org/md	Basic ox-hugo
Startup	Custom my-home buffer	Dashboard

3.3 Unique Features in Your Config

1. **Parallel L^AT_EX Preview** - 4 concurrent processes, SVG/PNG toggle
2. **Advanced Org-roam Dailies** - Recurring templates (daily, weekly, biweekly, monthly, yearly)
3. **Multi-account Email** - Gmail, UNSW, Abaj via notmuch/lieer/mb-sync
4. **Anki Integration** - Direct flashcard creation from org

5. **Inkscape Figures** - L^AT_EX-style figure insertion
6. **Hugo Blog Workflow** - org-to-markdown with custom ID handling
7. **Java LSP** - Performance-tuned jdtls

4 Refactoring Roadmap

4.1 Goal

Convert modular `elisp/*.el` structure to single `config.org` with org-babel tangling.

4.2 Phase 1: Infrastructure

4.2.1 Create New Directory Structure

```
~/.emacs.d/
├── init.el           # Simplified to 3 lines
├── early-init.el    # Keep as-is
├── config.org        # New main config
├── scripts/         # Move standalone modules
│   ├── ob-markdown.el # Pandoc integration
│   ├── ink.el        # Inkscape (has XML template)
│   ├── gruber-themes.el # Theme utilities
│   └── my-home.el    # Custom startup
└── elisp/
    └── custom-vars.el # Keep for customize system
```

4.2.2 New `init.el`

```
;;; init.el -*- lexical-binding: t; -*-
(setq custom-file (expand-file-name "elisp/custom-vars.el" user-emacs-directory))
(org-babel-load-file (expand-file-name "config.org" user-emacs-directory))
(when (file-exists-p custom-file) (load custom-file))
```

4.3 Phase 2: Section-by-Section Migration

Migrate in this order (low risk to high risk):

Order	Section	Source	Risk
1	Bootstrap	bootstrap.el	Low
2	Core packages	package-config.el (partial)	Low
3	UI & Appearance	ui-config.el	Low
4	Auto-save	auto-save-config.el	Low
5	Magit	magit-bindings.el	Low
6	Helm	package-config.el (partial)	Medium
7	Org Core	org-config.el (partial)	Medium
8	Org Babel	org-config.el (partial)	Medium
9	Org $\text{\LaTeX}$ Preview	org-config.el (partial)	High
10	Org-roam & Dailies	package-config.el (partial)	High
11	Email	email-config.el	Medium
12	Java LSP	java-lsp.el	Low
13	Anki	anki-config.el	Medium
14	Hugo	ox-hugo-bindings.el	Low

4.4 Phase 3: Testing Checklist

After migration, verify:

- ☐ Emacs starts without errors
- ☐ Helm M-x works with fuzzy matching
- ☐ `org-roam-node-find` works
- ☐ Daily capture creates proper structure
- ☐ $\text{\LaTeX}$ preview generates images (both SVG and PNG)
- ☐ TikZ preview in export blocks works
- ☐ notmuch opens and syncs all accounts
- ☐ Theme toggle (`C-c T t`) works
- ☐ Anki push creates cards
- ☐ Hugo export generates correct markdown
- ☐ Java files get LSP support

4.5 Key Decisions

Decision	Choice	Rationale
Package Manager	Keep straight.el	Already works, no benefit to Elpaca
Tangle Output	Single config.el	Simpler than multiple files
Large Modules	Move to scripts/ custom-vars.el	Keep config.org manageable Customize system expects it

4.6 Estimated Result

Component	Est. Lines
config.org	~6,500 (with prose)
scripts/*.el	~670 (unchanged)
init.el	3
early-init.el	67 (unchanged)
Total	~7,240

5 Quick Reference: Org-Babel Tangle Syntax

5.1 Default Tangle Header

Put in config.org header to tangle all blocks to config.el:

```
#+PROPERTY: header-args:emacs-lisp :tangle config.el :results silent
```

5.2 Per-Block Override

Don't tangle a specific block:

```
#+begin_src emacs-lisp :tangle no
;; This code won't be tangled
#+end_src
```

Tangle to a different file:

```
#+begin_src emacs-lisp :tangle "scripts/custom.el"
;; Goes to scripts/custom.el
#+end_src
```

5.3 Useful Org-Babel Settings

```
;; Preserve indentation when tangling
(setq org-src-preserve-indentation t)

;; Don't ask for confirmation when evaluating
(setq org-confirm-babel-evaluate nil)

;; org-tempo for quick block insertion (<s TAB)
(require 'org-tempo)
```